VOLLEY Gen6 — native coupled campaign verification and precision budgets

19 October 2026 · CNR-01

Purpose and configuration

CNR-01 independently propagates the prescribed host burns and finite-mass releases from the earlier four/twelve-payload construction. It then probes command errors and derives provisional metrology and thrust-time requirements. This is numerical verification and requirements development for a surrogate mission. Motor performance, delivered flight accuracy and hardware compatibility remain unvalidated.

Input	Declared basis
Reference orbit	500 km, 97.4°; fixed inertial +Z J2
Gravity	$\mu = 3.986004418\times10^{14}$ m³/s²; radius 6378137 m; $J2 = 1.08262668\times10^{-3}$
Manifest	Four: two 1U at 1.33 kg and two 3U at 4 kg; twelve: six of each
Release schedule	Nominal 0–3600 s; common assessment at 90000 s
Targets	Constructed 3–9 m/s prograde increments from reference-host states; not customer target states
Release inputs	Spring 2.5 m/s; VOLLEY imposed 3–6 m/s with host assistance
Host	300 kg dry surrogate; ideal Isp 220 s; device proxies 16/30 kg per four payloads
Reference reserves	10 kg for both four-release cases; 20 kg for both twelve-release cases

The source control law returns the host to reference positions and restores its final state. The force model is distinct from the earlier 450 km, shared-pole GMAT–Orekit benchmark. No Earth-orientation, drag or atmosphere accuracy is inferred from agreement here. The API epoch is a time origin for this time-independent surrogate, not a selected launch date.

Independent full-sequence calculation

The included event-export derivative retains the earlier source's gravity, targeting, rocket and recoil calculations. It adds explicit burn/release records and omits the later dense-output/clearance-reporting stage. The derivation patch and upstream source identities are included. Recomputed fuel amounts exactly reproduce the archived reference cases in this environment. The twelve-spring case with 10 kg reserve still fails at transfer 10; its 20 kg countercase completes.

Orekit 13.1.6 starts from **one initial host state**, propagates between every event, independently applies rocket mass loss and parent/child recoil, and propagates each released payload to the common epoch. It never resets a later host or payload arc to a source sample. The control vectors and targets

are shared inputs; the target construction and control optimization are not independently solved. SciPy and Hipparchus both use DOP853, with separate force and bookkeeping implementations.

For a relative release vector u, mass m and pre-release total M, the implementation uses v_child =v_host +(1−m/M)u and v_parent =v_host −(m/M)u. The comoving kinetic-energy identity is ½m(M−m)|u|²/M. Burn mass follows M_after =M_before exp(−|Δv|/(Isp g0)). These are point-mass, ideal-impulse relationships.

51 native cases cover five nominal baselines, two step refinements and 44 deterministic probes. The four matched comparisons are:

Case	Native fuel used [kg]	Maximum payload target difference [μm]	Final host difference [μm]
4 spring 10kg	4.355163	6.628	4.317
4 volley 3 6 10kg	1.328501	6.628	1.492
12 spring 20kg	15.352517	7.978	14.124
12 volley 3 6 20kg	4.049854	7.134	8.657

The largest nominal payload difference in these matched comparisons is 7.978 μm. Halving the maximum step from 60 to 30 s in the two twelve-release comparisons changes payload positions by up to 16.133 μm. Mass, momentum and comoving-energy residuals are retained in raw output and checked. The numerical consistency threshold is 1 cm position and 10^{-5} m/s velocity; these thresholds are not flight accuracy specifications or rigorous continuous error enclosures.

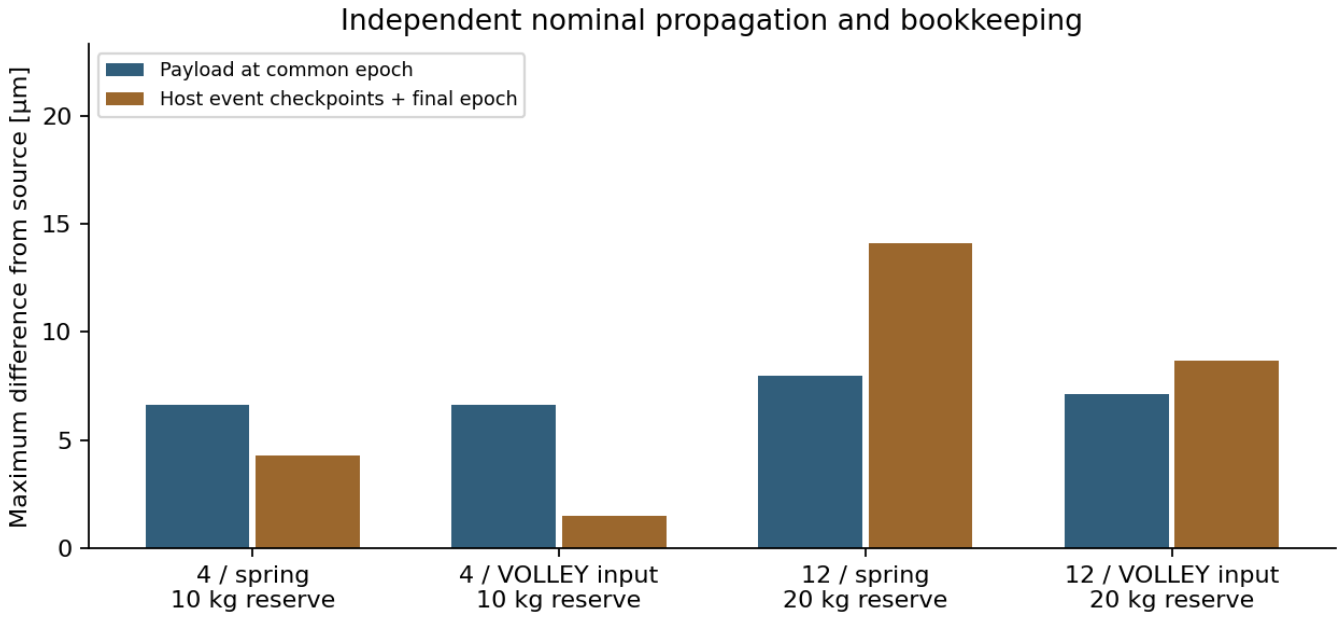


Figure 1: Independent nominal calculation; differences reflect shared model inputs

Command-error sensitivity

The probes vary one input at a time: common release-speed bias ±1% and +0.1%; in-plane/out-of-plane direction error ±0.1°; host-burn magnitude bias ±0.1%; and shot latency 5/50 ms. Every case uses the original commands, without navigation feedback or retargeting. Delayed shots also delay the following transfer burn, retaining the 20 s post-release coast. Assessment remains at the nominal common 90000 s epoch.

These are deterministic assumptions rather than measured distributions. The matrix reports the largest payload error across each stated signed probe, not a global uncertainty bound. Source-state differences at shifted intermediate events are not numerical residuals; nominal consistency and perturbed terminal errors are evaluated separately.

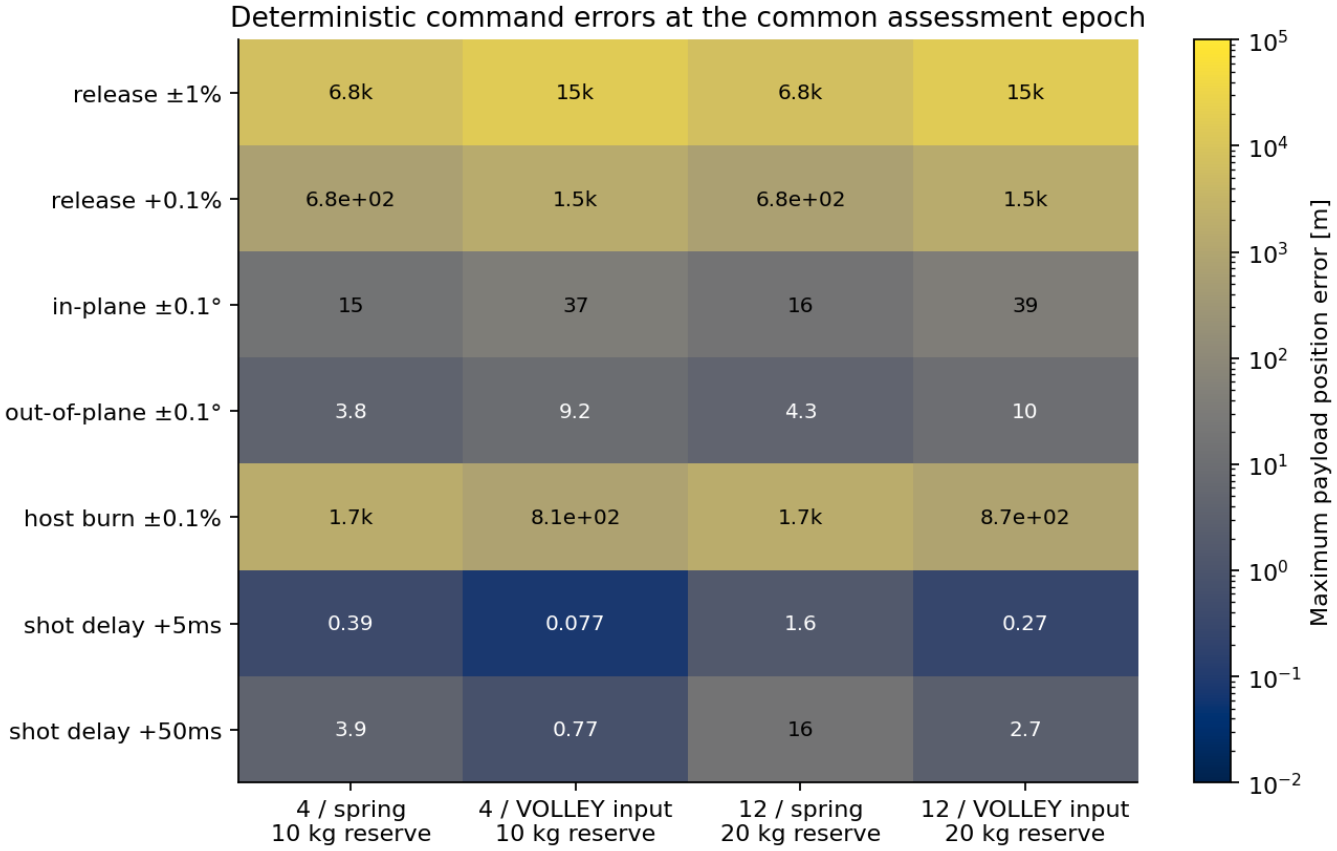


Figure 2: Maximum terminal position errors under explicitly defined probes

An illustrative 100 m / 0.1 m/s full-state box gives the following first-order allocation to a common release bias when **all other errors are set to zero**. It is not a customer requirement. The linearity ratio compares ten times the+0.1% position response with the+1% response; a ratio near one supports the local approximation over those probes.

Case	Common release-bias allocation	Linearity ratio
4 spring 10kg	0.01339%	0.99999
4 volley 3 6 10kg	0.00588%	1.00001
12 spring 20kg	0.01338%	0.99999
12 volley 3 6 20kg	0.00593%	1.00001

The 10 m / 0.01 m/s and 1000 m / 1 m/s alternatives are in SUMMARY.json. Their purpose is to connect mission targets to actuator and measurement requirements. A controller must estimate the host state, account for actual pointing and manage release velocity. Published model-versus-test error in another deployer is not a measurement of VOLLEY repeatability.

Velocity measurement budget

For two photogates with baseline L and transit time t, the constant-speed estimate is L/t. The 324 static intervals use speeds 3/6/9 m/s, baselines 50/100/200 mm, calibration bounds 5/10/50 µm, indi-

vidual timestamp bounds 0.1/1/10 µs and residual temperature bounds 0/0.1/1/10 K. Aluminium-like CTE 23×10^{-6}/K is assumed. Nine m/s remains a study point beyond the first-article objective.

The exact constant-speed interval is $(L-\delta L)/(t+2\delta t)$ to $(L+\delta L)/(t-2\delta t)$. Temperature drift adds CTE×L×δT to the distance bound. These are prescribed bounds, not statistical standard deviations or actual detector specifications.

A further 108 cases include mean-to-exit velocity and frame compensation. With |a_relative|≤a and positive monotonic motion, the transit-time comparison bounds are$2L/(v0+\sqrt{(v0^2\pm2aL)})$. The difference between mean transit speed and exit speed is bounded by a×t_max/2. A rate-knowledge bound adds δω×r for a 0.5 m lever arm. Exact constant-acceleration corner checks exercise the interval arithmetic.

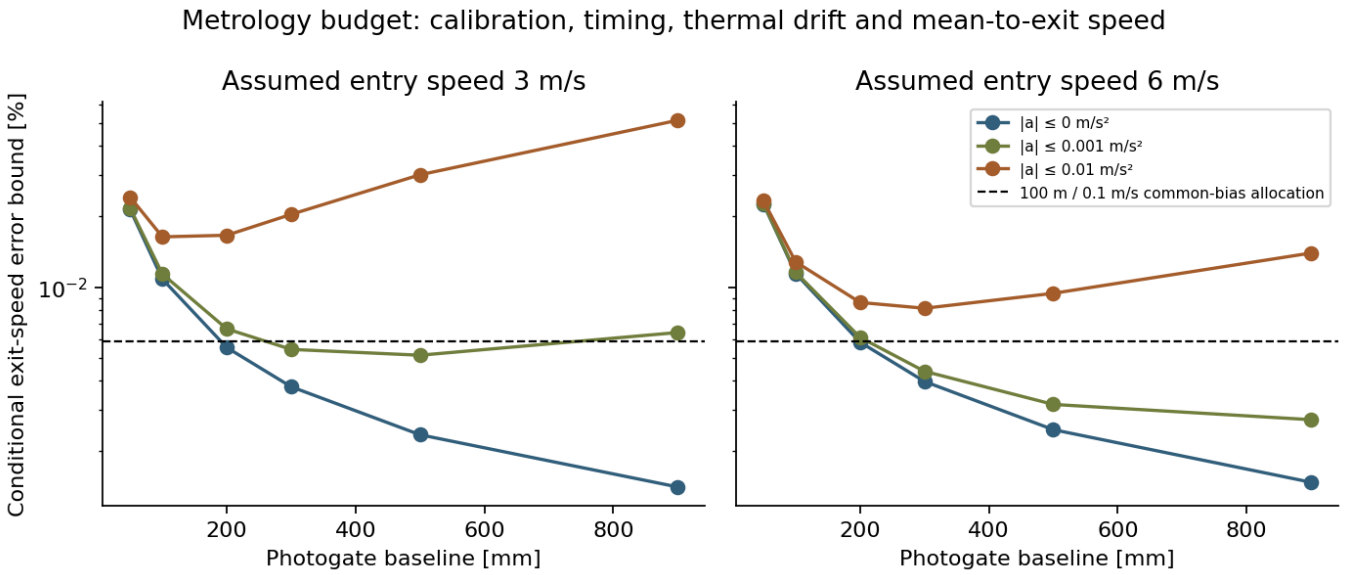


Figure 3: Conditional measurement/exit-speed budget at 3 and 6 m/s; dashed allocation consumes the entire illustrative mission budget

A longer baseline reduces calibration/timestamp contributions but increases the mean-to-exit contribution when relative acceleration is present. The required quiet/coasting measurement zone, baseline thermal calibration, detector response, payload edge geometry, tip-off and frame-rate compensation must be implemented and tested. Post-release measurements support verification and calibration; they cannot correct a passive satellite already released. Closed-loop acceleration and release control require their own sensors and design.

Necessary thrust-time screen

For an equivalent constant-thrust engine, the nominal impulse fuel amount implies thrust-on time Δm×Isp×g0/F. Twenty cases use 0.1/1/5/20/100 N. This is a necessary time calculation for the **frozen reference commands**; it does not integrate finite burns or optimize another spring/VOLLEY policy.

Case at 5 N	Thrust-on time before final release [s]	Total-time screen
4 spring 10kg	1432.7	Not excluded by total time alone
4 volley 3 6 10kg	356.4	Not excluded by total time alone
12 spring 20kg	6134.9	Exceeds reference window
12 volley 3 6 20kg	1487.6	Not excluded by total time alone

Passing the summed-time screen is insufficient: each finite burn must fit the chronology and avoid incompatible operations. Alternative controls/release epochs can change these demands. A low-thrust

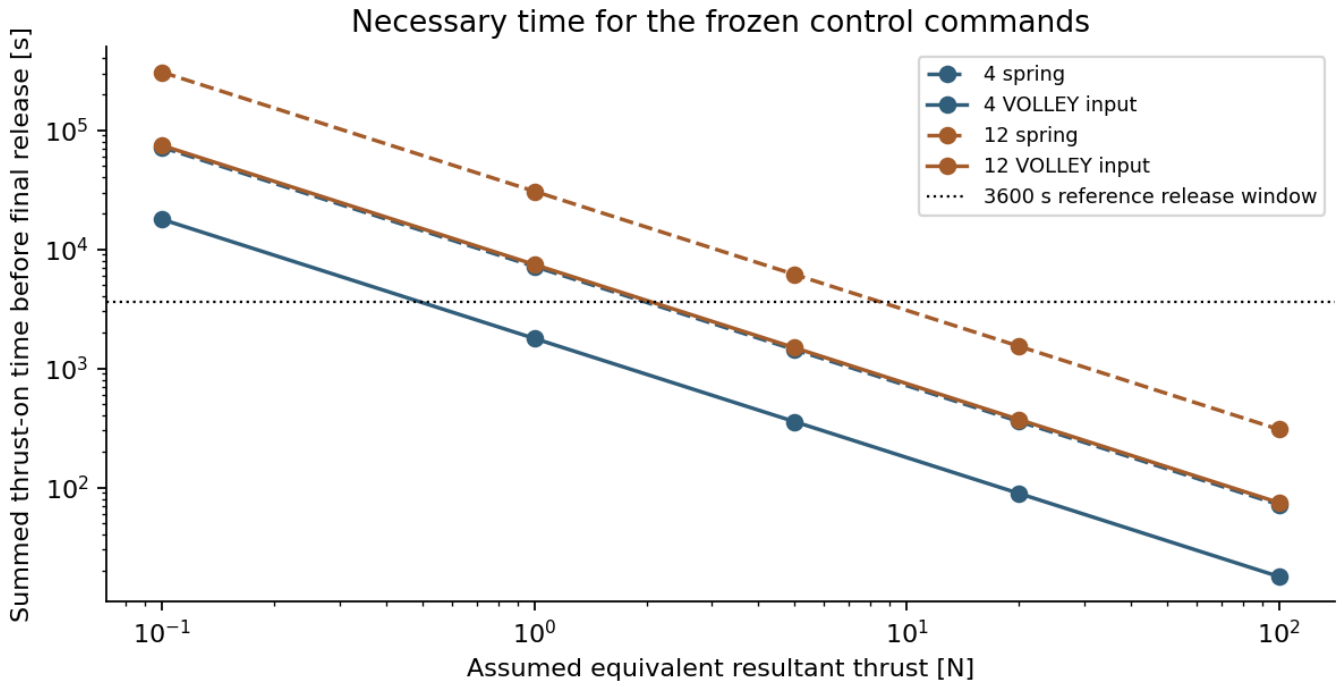


Figure 4: Frozen-command thrust-on time; coasting, feeding, slew and safety scheduling need additional allowance

failure here does not prove every competent spring-host strategy fails. The device mass proxies still leave VOLLEY heavier despite lower fuel consumption. A selected host, installed system and optimized service comparison remain necessary before a product advantage is claimed.

Verification status and remaining engineering

Verified within this study	Still required
Prescribed full host/release sequence in independent native code	Customer target states and acceptable errors
Rocket/recoil/mass/energy identities and step comparison	Finite-burn targeting, attitude/slew and navigation feedback
Deterministic terminal-error responses	Calibrated sensor/actuator/error distributions
Conditional metrology and total-thrust-time calculations	Actual measurement architecture and source-rated hardware
Retained failed low-reserve spring campaign	Body/contact/near-field clearance and fault recovery

The earlier conditional point-centre clearance screen remains conditional. Numerical agreement at events and the common epoch does not prove the assumed 1 cm continuous state-error enclosure, physical separation geometry or conjunction probability. All 51 trajectories use imposed release velocities; they do not establish a motor rating.

Reproduction

1. Inspect the reports, raw event/host/payload CSVs and exact source identities.
2. Run check_capture.py with standard Python for integrity and independent bookkeeping checks.

3. Run download_jars.py to obtain the pinned public Orekit/Hipparchus dependencies; set OREKIT_CLASSPATH to jars/* and use Java 21 with its source-file compiler module.
4. Run run_full_campaign.py for the native cases, then summarize.py and report.py for budgets and figures. Install the pinned numerical packages when regenerating the source events with instrument_campaign.py and export_events.py.
5. Compile this Markdown with pandoc and XeLaTeX for the PDF. The manifest records the captured environment and exact inputs/outputs; rerunning does not imply bit-identical PDF metadata.